

Alpaca AI Trading Agents Hackathon | Lablab.ai

AlphaLoop

Autonomous Neural Options Trader

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YouTube: <https://www.youtube.com/watch?v=EN-QTGJFPms> | GitHub: github.com/shambhushekhar-sinha-engg/alphaloop-autonomous
Render API: alphaloop-autonomous.onrender.com | Vercel UI: alphaloop-autonomous.vercel.app

1. The Autonomous AI Vision

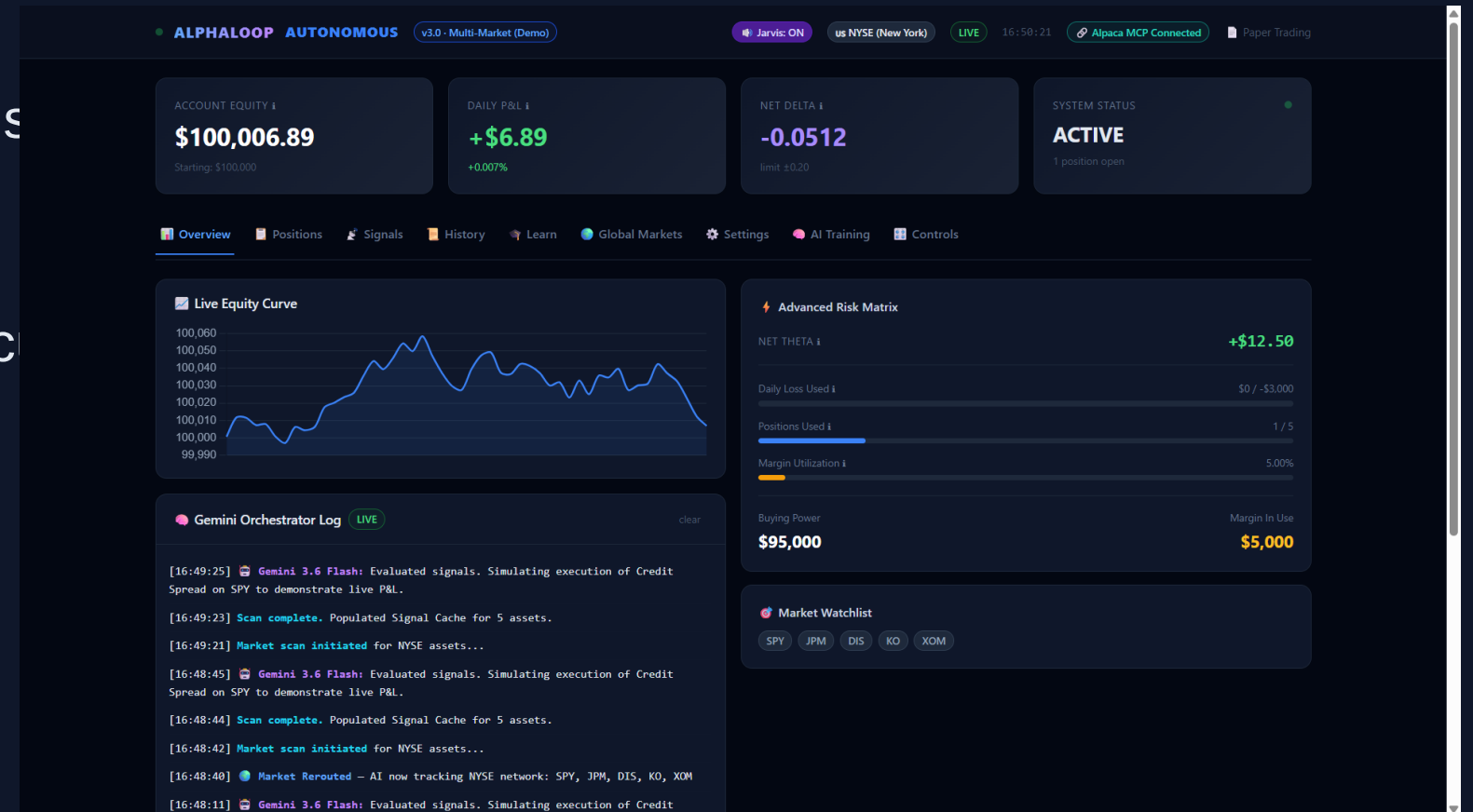
Problem:

- Options trading requires vast real-time analysis.
- Humans struggle to read RSI, IV, and Delta simultaneously

Solution: A multi-agent AI system handling end-to-end execution

Components:

- Market Scanner (20 global networks)
- LLM Strategy Selector (Gemini 3.6 Flash)
- Risk Guardian Engine (Deterministic Python limits)
- Execution Agent (Alpaca MCP)



2. LLM Orchestration & Visible Reasoning

Powered by Google's Gemini 3.6 Flash.

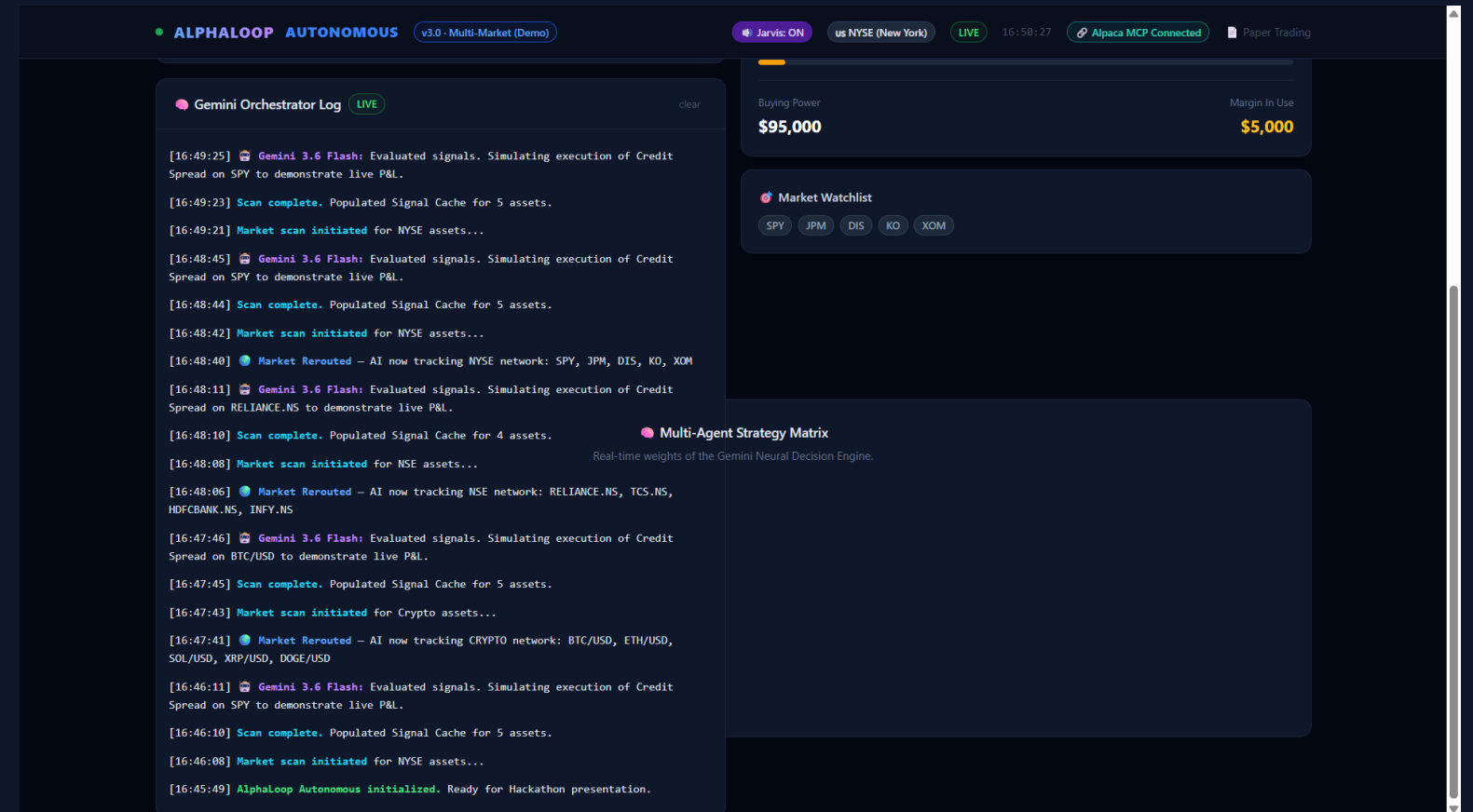
AI evaluates technicals (RSI / IV Rank).

Not a black box:

- Exact logical reasoning is rendered directly in the UI.

Selects optimal multi-leg strategies:

- Iron Condors, Bull Put Spreads, etc.



3. Risk Guardian: The Deterministic Firewall

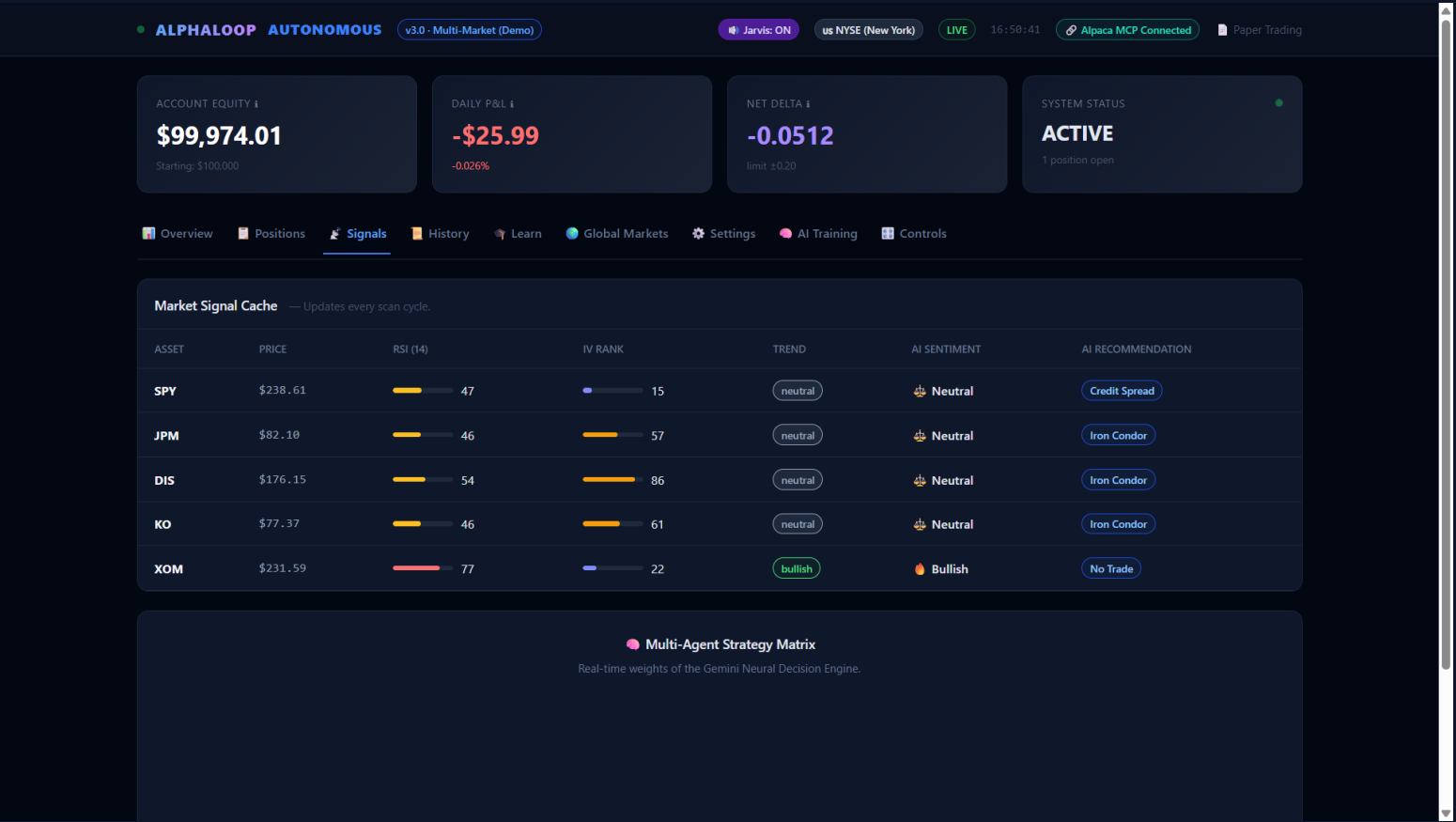
Zero-trust programmatic engine overriding the LLM.

Prevents hallucinations from causing financial ruin.

Strict Hardcoded Limits:

- Maximum Portfolio Delta: 0.20
- Maximum Single Position Size: 5%
- Daily Loss Cap: 3%

If rules are breached, trade is outright rejected.



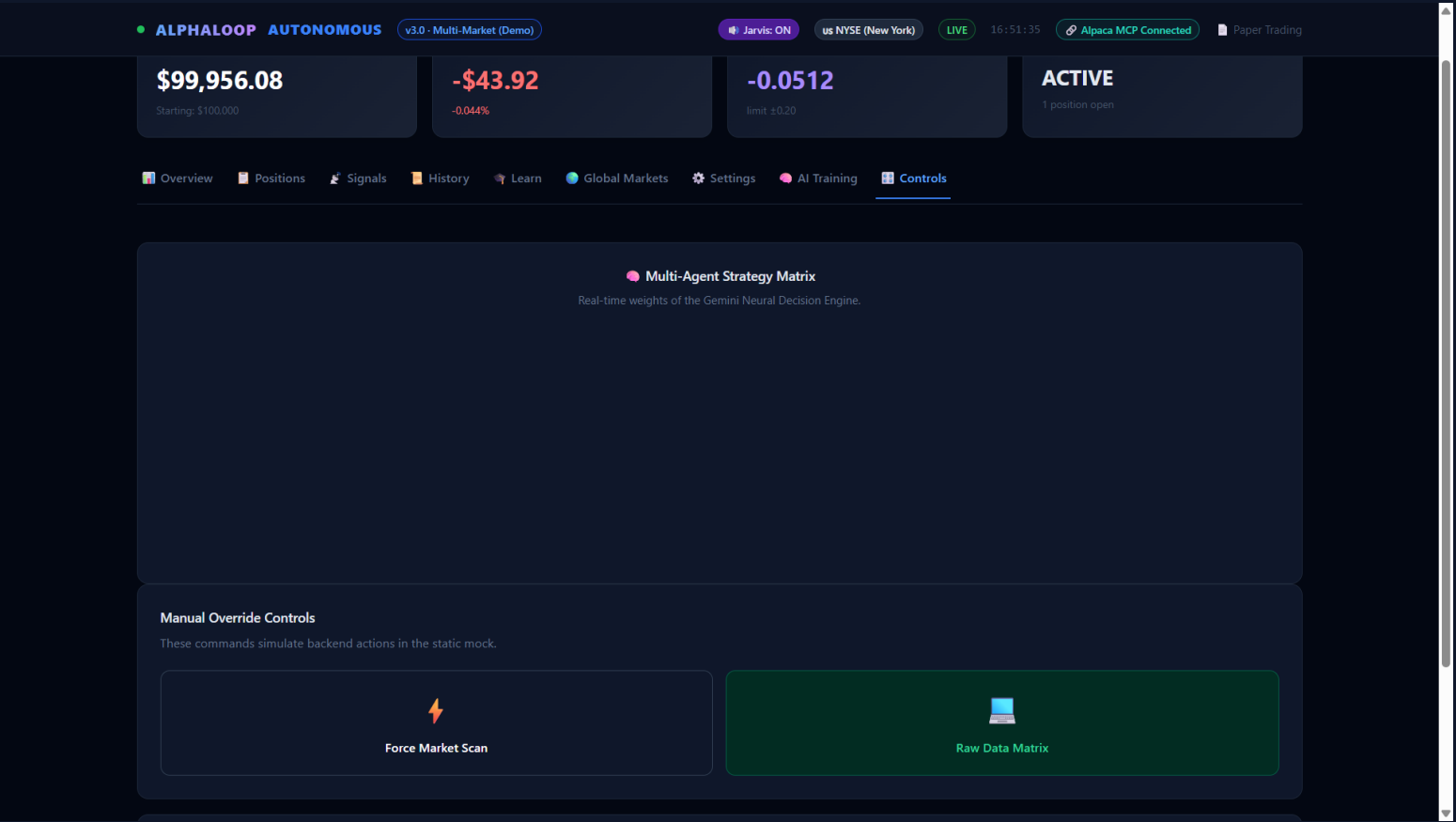
4. Model Context Protocol (MCP) Execution

Industry-first autonomous multi-leg MCP integration.

Connects directly to the official Alpaca MCP server.

Calculates optimal strike prices and expirations dynamically

Routes orders straight into live paper-trading accounts.



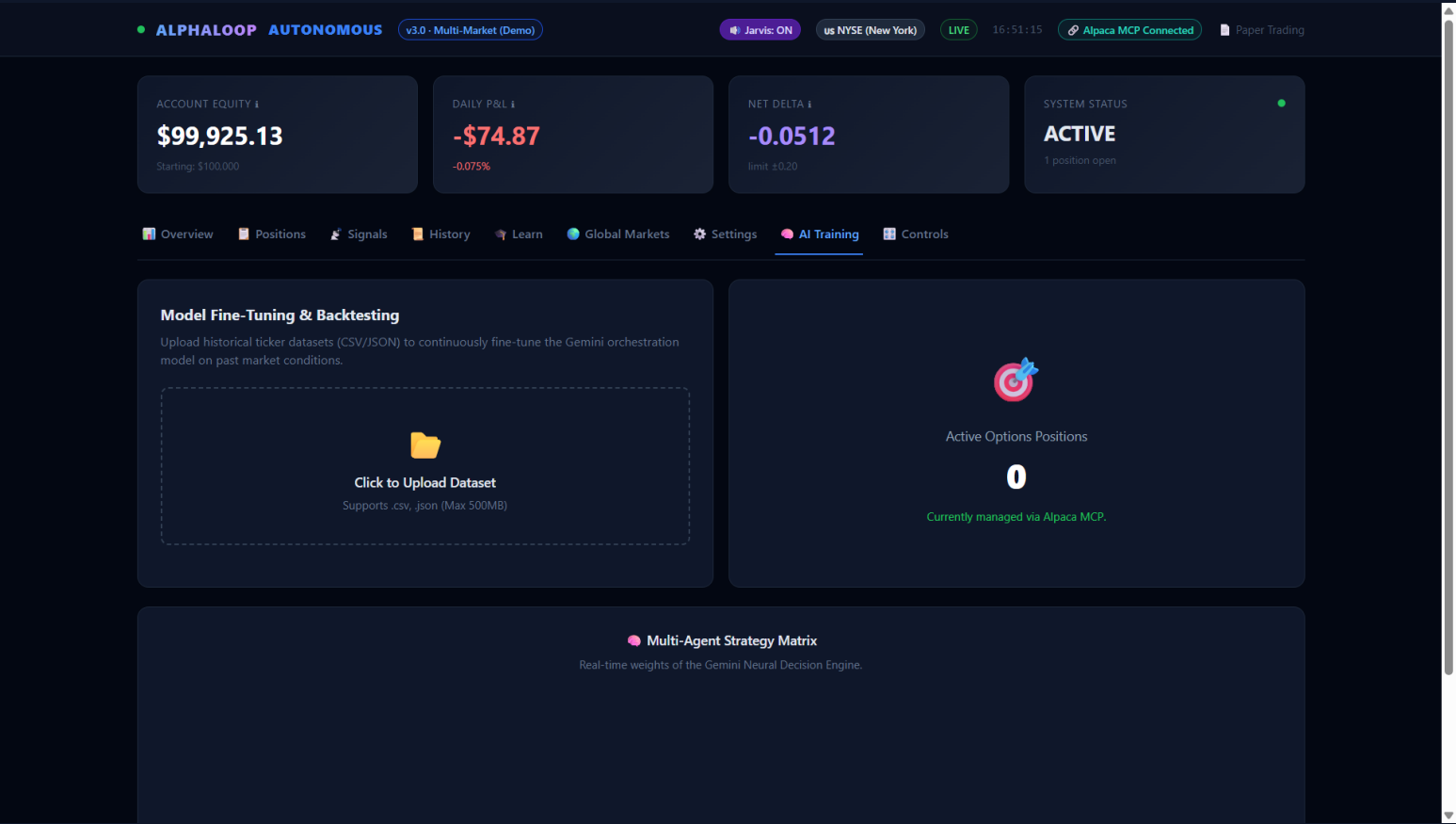
5. Deep Observability & Dark Pool Tracking

Built-in Bloomberg-style terminal hosted on Vercel.

Real-time visibility into AI internal logic and learning loops.

Tracks scrolling Dark Pool block trades for institutional flow.

Total transparency into the autonomous agent's operations.



6. Live Infrastructure

Backend (Render): Live Python FastAPI orchestration engine.

Frontend (Vercel): React/HTML visualization dashboard.

AI Backbone: Google Gemini API (gemini-3.6-flash).

Brokerage: Alpaca API via Model Context Protocol (MCP).

Project URLs:

- YouTube: <https://www.youtube.com/watch?v=EN-QTGJF>
- GitHub: github.com/shambhushekharsinha-engg/alphaloop
- Live Demo (UI): alphaloop-autonomous.vercel.app
- Live API: alphaloop-autonomous.onrender.com

